Annexure – IV

Proforma for Outline of Research Work (ORW)

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6. Date of joining	:	03.10.2024
7. Title of the research project	:	Corporate Adoption of Blue Carbon Credit Mechanism: An Economic Feasibility and Strategic Impact Analysis
8. Objective	:	1. To analyse the relationship between corporate carbon emissions and the adoption of blue carbon credit mechanisms among Indian firms. 2. To evaluate the economic feasibility of investing in blue carbon credits from the perspective of corporate buyers. 3. To assess the strategic impact of adopting blue carbon credit on corporate ESG performance and investor perceptions.
9. Duration	:	110 Days
10. Location	:	Chennai

11. Introduction/ Background of research

Climate change has become one of the most pressing environmental issues today, largely resulting from human activity since the Industrial Revolution. The rise in greenhouse gas emissions from industrial and economic development has contributed to higher global temperatures and more frequent extreme weather events. While natural processes also influence the climate, human-induced greenhouse effects are now a key driver of global warming (Mikhaylov et al., 2020).

In this context, Over the past two decades, corporations have become influential actors in climate politics. Although many large companies publicly commit to UN-aligned "net-zero" targets and position themselves as leaders in addressing climate change through technological innovation, they often oppose strong carbon regulation while continuing practices that contribute to emissions. Consequently, corporations play a significant role in shaping both public discourse and policy responses to climate change in ways that frequently prioritize profit and economic growth over substantive emissions reductions (Wright & Nyberg., 2024). Greenhouse gas emissions create a negative externality because polluters do not bear their full social costs. To address this, governments use carbon

pricing—mainly through carbon taxes or emissions trading schemes (ETS)—to internalize these costs under the "polluter pays" principle (Ji et al., 2018).

Blue carbon ecosystems mangroves, tidal marshes, and seagrasses—store large amounts of carbon and provide key benefits such as coastal protection and biodiversity support, but they have been heavily degraded by human activity. Blue carbon projects (BCPs) use voluntary carbon markets to finance the restoration and protection of these ecosystems by generating tradable carbon credits. Despite credibility problems in other nature-based carbon markets, BCPs are growing and attracting higher prices. This study assesses existing BCPs and finds that they are already contributing to climate mitigation and biodiversity goals through financial flows from developed to developing countries (Farahmand et al., 2025).

This research seeks to evaluate the economic feasibility, financial returns, and business benefits of corporate investment in blue carbon projects while also analysing how these credits can strengthen competitive advantage, brand reputation, and long-term resilience in fisheries and aquaculture. By integrating carbon finance with corporate strategy and coastal ecosystem management, the study aims to propose a viable business model that supports sustainable resource use, benefits coastal communities, and aligns corporate growth with global climate objectives.

12. Review of work done

Mohajan., (2017) Global warming is primarily caused by human-generated greenhouse gas emissions, especially from burning fossil fuels such as coal, oil, and gas. Since the Industrial Revolution, atmospheric CO_2 levels have risen from 280 ppm to about 450 ppm, resulting in a global temperature increase of nearly 0.8°C over the last century. Major emitters like the USA and China contribute a large share of global emissions. Although international efforts such as the Kyoto Protocol and climate summits have been made, they have not been sufficient to control emissions. As a result, the world is experiencing ocean warming, rising sea levels, and increasing risks to water supply, food security, and human health, making urgent climate action essential.

Rahman et al., (2025) Among 12 MENA countries, Saudi Arabia, Egypt, Algeria, United Arab Emirates, and Qatar were the five largest contributors to greenhouse gas (GHG) emissions, with Saudi Arabia consistently leading. Saudi Arabia's emissions increased significantly from 2.50×10^8 t CO_2eq in 1990 to 8.79×10^8 t CO_2eq by 2023. Egypt also showed a steady upward trend, rising from 1.41×10^8 to 3.68×10^8 t CO_2eq over the same period. The United Arab Emirates experienced the sharpest annual surge, with a 33% increase between 1999 and 2000. Overall, the energy sector was the primary driver of emission growth across the region. Total GHG emissions in the 12 MENA countries expanded from 5.70×10^8 t CO_2eq in 1990 to 14.78×10^8 t CO_2eq in 2019. By 2019, the top five emitting countries accounted for nearly 91% of the region's energy-related emissions, with Saudi Arabia alone contributing about 36%.

Wright & Nyberg., (2024) Corporations have become key actors in climate politics, publicly committing to "net-zero" targets while actively shaping policy to favour business-led, market-based solutions over strict government regulation. Despite this green rhetoric,

major corporations, especially fossil fuel companies, have continued to expand fossil fuel production and remain major drivers of emissions. Businesses promote technological innovation, self-regulation, and carbon markets as the primary solutions, while lobbying against stronger regulations. In response to growing public pressure and the Paris Agreement, many companies have adopted net-zero language and invested in renewables and new technologies. Yet, their core profit-driven model and reliance on fossil fuels continue to challenge meaningful climate action.

Meena., (2015) Carbon credits are tradable permits that represent the right to emit one ton of carbon dioxide (or its equivalent). The system originated from the 1997 Kyoto Protocol to control greenhouse gas emissions by putting a price on pollution. Governments set emission limits, and companies that exceed them can buy credits from those that reduce or remove emissions through activities like renewable energy projects or tree planting. The system aims to incentivize emission reductions through market-based trading. Carbon credits are classified into Carbon Offset Credits (clean energy projects) and Carbon Reduction Credits (carbon capture and storage through nature-based methods).

Gupta., (2011) Carbon trading is a market-based mechanism created under international climate agreements to control greenhouse gas emissions. Governments set emission quotas for countries and businesses, and carbon credits allow one tonne of CO_2 (or equivalent GHG) to be emitted. Firms that exceed their limits must buy credits, while those that emit less can sell their surplus in global markets such as the Chicago and European Climate Exchanges. This system encourages companies to adopt cleaner technologies, reduce emissions, and earn profits through selling credits. Individuals can also buy credits to offset their carbon footprint. India has become a major player in the carbon credit market due to its low emissions and numerous Clean Development Mechanism projects, benefiting companies, renewable energy firms, farmers, and initiatives like the Delhi Metro. Overall, carbon trading promotes environmental sustainability while creating economic incentives for greener practices.

Friess et al., (2022) BCEs store large amounts of carbon in waterlogged soils, sequester CO_2 efficiently, and can potentially offset significant emissions globally. Despite their high climate and co-benefits, including fisheries, coastal protection, and livelihoods, only a few projects currently generate carbon credits. Scaling conservation and restoration of BCEs could avoid hundreds of Tg CO_2e annually and provide economic value exceeding $190 billion per year, making them a promising complement to decarbonization and corporate net-zero strategies.

Chowdhury et al., (2018) Coastal regions, though only 4% of the world's land, host one-third of the global population and are highly vulnerable to climate change impacts like sea level rise and erosion. Mangroves are among the most carbon-rich ecosystems, storing 49–98% of their carbon in anoxic soils and acting as highly effective blue carbon sinks. Restoration and conservation of mangroves, particularly through community-based programs in the Indian Sundarbans, provide dual benefits: disaster protection and carbon sequestration under REDD+ initiatives. Effective mangrove species selection, planting

methods, and monitoring of soil organic carbon are crucial for maximising cost-effective carbon sequestration and sustaining ecosystem services.

Friess et al., (2022) Corporate strategy increasingly incorporates sustainability, particularly through blue carbon initiatives, which focus on the carbon sequestration potential of coastal ecosystems like mangroves and seagrasses. These ecosystems can sequester carbon at rates exceeding those of terrestrial forests, presenting significant opportunities for companies aiming to offset emissions and achieve net-zero targets.

(Friess et al., 2022; Thomas, 2015) However, the integration of blue carbon into corporate strategies faces challenges, including limited supply of blue carbon credits and the complexities of verifying carbon storage in these ecosystems.

Macreadie et al., (2017) Effective management practices, such as reducing nutrient inputs and restoring hydrology, are essential for optimizing blue carbon sequestration.

(Haque & Ntim, 2022; Charnovitz & Esty, 2013) Furthermore, corporate sustainability initiatives have been shown to enhance carbon performance, particularly in high-emission industries, suggesting that a commitment to environmental sustainability can drive both ecological and economic benefits. Thus, blue carbon represents a critical intersection of corporate strategy and sustainability, necessitating innovative approaches to resource management and market engagement.

13. Rationale

The growing interest in blue carbon ecosystems such as mangroves, seagrasses, salt marshes, and seaweed has created new opportunities for corporations to align climate action with business value. In fisheries and aquaculture, these ecosystems not only support environmental sustainability but also present potential financial opportunities through carbon credits, green investments, and market differentiation. However, uncertainties regarding costs, returns, and practical implementation have limited corporate participation. This study is therefore important in evaluating whether investments in blue carbon projects can be economically feasible and financially beneficial for corporations while contributing to climate mitigation.

Developing sustainable business models around blue carbon requires a clear understanding of revenue streams, investment risks, and long-term economic viability. By analysing the financial benefits, cost structures, and business opportunities associated with blue carbon initiatives such as mangrove restoration and seaweed farming, this study aims to provide practical insights for integrating blue carbon mechanisms into profitable and sustainable corporate strategies within fisheries and aquaculture value chains.

14. Broad outlines of work with methodology

Objective 1: To analyse the relationship between corporate carbon emissions and the adoption of blue carbon credit mechanisms among Indian firms

A list of Indian companies publishing Sustainability / BRSR / ESG reports will be prepared

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Companies disclosing carbon emission data (Scope 1 and Scope 2 emissions in tCO_2e) will be selected

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Data on total annual carbon emissions will be extracted from sustainability reports

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Information on blue carbon credit purchase (volume of credits purchased and investment amount) will be recorded by questionnaire

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A Carbon Offset Ratio will be calculated using:

$$Carbon\ Offset\ Ratio = \frac{Blue\ Carbon\ Credit\ Purchased}{Total\ Carbon\ Emissions}$$

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Descriptive statistics (mean, standard deviation, percentage adoption rate) will be computed

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Pearson correlation analysis will be conducted to examine the relationship between emission levels and credit adoption

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Linear regression analysis will be performed to determine whether emission levels significantly influence blue carbon adoption

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Sector-wise comparison (e.g., manufacturing, energy, IT, Fisheries) will be conducted to identify variation in adoption patterns

Objective 2: To evaluate the economic feasibility of investing in blue carbon credits from the perspective of corporate buyers

Data on the cost of purchasing carbon credits will be compiled from corporations by questionnaire

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Investment values in blue carbon credits will be estimated for selected firms

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Total Investment = Carbon Credits Purchased × Market Price per Credit

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Potential financial benefits, such as compliance cost reduction, ESG-related capital access, and reputational advantage, will be identified

$$WACC = \left(\frac{E}{v} x R_e\right) + \left(\frac{D}{v} \times R_d \times (1 - T_C)\right)$$

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A Cost–Benefit Analysis (CBA) will be performed by comparing the total investment cost with the estimated financial benefits

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Net Present Value (NPV) will be calculated using projected cash flow benefits:

$$NPV = \sum_{t=1}^{n} \frac{R_t}{(1+i)^t} - C_0$$

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Internal Rate of Return (IRR) will be computed using Excel financial functions

$$0 = NPV = \sum_{t=0}^{n} \frac{CF_t}{(1+r)^t}$$

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The Benefit–Cost Ratio will be calculated to determine financial viability

$$BCR = \frac{Present\ Value\ of\ Benefits}{Present\ Value\ of\ Costs}$$

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Sensitivity analysis will be conducted by varying carbon credit prices and discount rates

Objective 3: To assess the strategic impact of adopting blue carbon credit on corporate ESG performance and investor perceptions

A structured questionnaire will be designed

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The questionnaire will measure:

1. ESG performance enhancement
2. Investor confidence improvement
3. Brand value impact
4. Competitive positioning

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The survey will be distributed to sustainability managers, ESG officers, CSR heads, and corporate finance professionals

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Reliability of the instrument will be tested

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Exploratory Factor Analysis will be conducted to group strategic impact indicators

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Regression analysis will be performed to examine the influence of blue carbon adoption on ESG perception and investor confidence

15. Expected outcomes

1. Corporate Carbon Emissions and Blue Carbon Credit Adoption:
The study is expected to determine whether companies with higher carbon emissions are more likely to adopt blue carbon credits. It may reveal sector-wise differences in adoption patterns. The findings will clarify how much emissions are actually being offset.

2. Economic Feasibility of Investing in Blue Carbon Credits:
The research is expected to evaluate whether investing in blue carbon credits is financially viable. It may show whether the investment generates positive returns (NPV and IRR). The study will also identify key financial risks and cost factors.

3. Strategic Impact on Corporate ESG Performance:
The study is expected to assess whether blue carbon adoption improves ESG performance and investor confidence. It may highlight its impact on brand value and competitive advantage. The findings will indicate whether it creates long-term strategic benefits.

16. Research Paper to Be Published and Tentative Title

Corporate Adoption of Blue Carbon Credit Mechanism: An Economic Feasibility and Strategic Impact Analysis. (Indian Journal of Economics and Development, NAAS 6.20)

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Signature of the Student